

RICE (INDRAYANI, BASMATI & KOLAM)

Complete Deep Farming, Grading & Traceability Report

Maharashtra Edition

Districts Covered: Pune · Kolhapur · Satara · Solapur · Nashik · Nagpur · Chandrapur · Gondia ·
Bhandara · Raigad · Ratnagiri · Sindhudurg

*Compiled from ICAR-Indian Institute of Rice Research (IIRR), Government of India Official Records,
Maharashtra State Agriculture Department & Millets Now Traceability Platform V2*

Version: 2.0 | June 2026

Millets
Now

1. INTRODUCTION & ORIGIN OF RICE

Rice, botanically known as *Oryza sativa* L., is one of the world's oldest domesticated cereals and the central staple food for a large share of the global population. Asian rice evolved through long selection under wetland, rainfed, upland and irrigated systems, creating thousands of grain types, aroma classes and cooking qualities.

In Maharashtra, rice is grown under sharply different ecologies: high-rainfall Konkan lowlands, Western Ghats valleys, assured-irrigation pockets, and eastern Vidarbha paddy belts. The report focuses on Indrayani, Basmati and Kolam because they represent premium aroma, long-grain and everyday fine-grain markets.

1.1 Global & National Standing

Rice is the leading food grain across Asia and a strategic commodity in public distribution, nutrition programmes, exports and domestic retail. India is among the world's largest rice producers and exporters, with both non-basmati and basmati supply chains requiring strong varietal identity and quality assurance.

Maharashtra is not the largest Indian rice state, but it has high-value regional niches: Ambemohar/Indrayani aroma traditions near Pune and Western Ghats, fine Kolam in Vidarbha and urban retail, and selected Basmati-style long-grain cultivation where soil, temperature and market contracts support it.

1.2 Why Rice Matters

- Food security: rice is a dependable staple across rural and urban households.
- Premium markets: aroma, grain length, whiteness, milling recovery and purity create strong price differentiation.
- Agro-ecological fit: paddy suits high-rainfall Konkan and Vidarbha lowlands where water is seasonally abundant.
- Processing value: rice supports milling, poha, flakes, flour, bran oil, starch and snack industries.
- Traceability need: premium varieties are vulnerable to admixture, so lot-level identity improves buyer confidence.

1.3 Agronomic Significance in Maharashtra

Rice is concentrated in Konkan districts, eastern Vidarbha and the irrigated valleys of western Maharashtra. Districts such as Raigad, Ratnagiri and Sindhudurg rely on monsoon water and banded fields, while Gondia, Bhandara and Chandrapur have strong paddy culture with tanks, canals and lowland clay soils.

Pune, Kolhapur and Satara contribute specialty and regional-market rice, including aromatic Indrayani zones. Nashik and Solapur are more selective rice districts, suitable mainly where irrigation, soil puddling and market linkage justify the crop.

Rice Name	Primary Market Identity	Maharashtra Position	Traceability Priority
Indrayani	Short/medium aromatic, soft-cooking	Premium regional rice in Pune and adjoining Ghats	Aroma identity and local origin
Basmati	Extra-long slender aromatic grain	Contract/specialty, not a universal fit	Varietal purity and elongation
Kolam	Fine, lightweight, non-sticky everyday rice	Strong retail and Vidarbha/Konkan demand	Whiteness, polish and broken control

Millets
Now

2. RICE VARIETIES OVERVIEW

2.1 Indrayani Rice

Indrayani is a well-known aromatic rice associated with Maharashtra's Western Ghats and Maval-Pune belt. It is valued for fragrance, soft cooked texture and regional food preference rather than maximum yield alone.

2.2 Basmati Rice

Basmati is a premium long-grain aromatic class recognised for kernel elongation after cooking, slender grain shape and distinctive aroma. Authentic Basmati trade requires strong varietal purity and standards-based grading.

2.3 Kolam Rice

Kolam is a popular fine rice type in Maharashtra retail markets. It is generally preferred for daily cooking because of fine grain, moderate cooking softness and affordability compared with premium aromatic rice.

2.4 Variety Comparison Matrix

Parameter	Indrayani	Basmati	Kolam
Grain type	Short to medium, aromatic	Extra-long slender aromatic	Fine to medium slender
Cooking quality	Soft, fragrant, slightly sticky	Long separate grains after cooking	Light, fine and household-friendly
Maturity tendency	Medium duration	Medium to long duration	Medium duration
Market positioning	Regional premium	Premium/export-style	Mass premium and daily retail
Key risk	Aroma dilution and admixture	Purity, grain breakage and elongation failure	Broken percentage and polish inconsistency

2.5 District-wise Suitability

District Group	Best Fit	Reason
Pune, Satara, Kolhapur	Indrayani, Kolam	Aromatic regional preference, valleys and assured monsoon/irrigation pockets.
Gondia, Bhandara, Chandrapur, Nagpur	Kolam, fine rice, selected long-grain lots	Traditional paddy belt with lowland soils and milling networks.
Raigad, Ratnagiri, Sindhudurg	Kolam, local fine rice, selected Indrayani-style lots	High rainfall Konkan fields, bunded lowlands and local consumption.
Nashik, Solapur	Contract Kolam or specialty irrigated rice	Rice only where water and market premium justify cost.

3. NUTRITIONAL PROFILE & HEALTH BENEFITS

3.1 Nutritional Composition per 100 g

Nutrient	White Rice	Brown/Unpolished Rice	Remark
Energy	345–365 kcal	350–370 kcal	Staple energy source.
Carbohydrates	77–80 g	74–77 g	Mostly starch; cooking quality depends on amylose.
Protein	6.5–8.0 g	7.0–8.5 g	Moderate cereal protein.
Fat	0.4–0.8 g	2.0–3.0 g	Bran layer carries more oil.
Dietary fibre	0.5–1.0 g	2.5–4.0 g	Higher in brown rice.
Iron	0.2–1.5 mg	1.0–2.0 mg	Depends on variety and processing.
Thiamine/B vitamins	Lower after polishing	Higher	Bran removal lowers B vitamins.

3.2 Glycemic Index & Dietary Benefits

Rice glycemic response depends on variety, amylose content, polishing, cooking method, cooling/reheating and meal pairing. Whole grain or parboiled forms generally improve satiety compared with highly polished soft rice.

3.3 Health Benefits

- Easily digestible staple suitable for children, elderly consumers and convalescent diets.
- Naturally gluten-free grain for wheat-free diets.
- Brown and hand-pounded rice retain more fibre, minerals and B vitamins.
- Parboiled rice improves milling recovery and retains more nutrients than highly polished raw rice.
- Rice bran supports value addition through edible oil and nutraceutical ingredients.

4. TYPES OF SEEDS & SEED SELECTION

4.1 Certified Seeds

Certified seed provides varietal identity, physical purity, germination assurance and lower risk of seed-borne disease. It is strongly preferred for traceable Indrayani, Basmati and Kolam supply chains.

4.2 Hybrid Seeds

Hybrid rice can offer high yield in suitable areas, but premium aroma and local cooking-quality markets often prefer named varieties. Hybrid use should be decided by market requirement, not yield alone.

4.3 Seed Selection Guidelines

Selection Parameter	Recommended Standard	Why It Matters
Germination	85% or higher	Ensures uniform nursery and field population.
Physical purity	98% or higher	Reduces off-type plants and grain admixture.
Moisture in seed	Safe dry seed from reliable storage	Maintains viability and avoids fungal risk.
Seed source	Certified agency, university, KVK or trusted supplier	Supports audit and traceability.
Variety identity	Packet, invoice and lot number retained	Required for premium rice branding.

Millet
Now

5. SEED VARIETY DEEP ANALYSIS

5.1 Indrayani Variety

Indrayani should be grown where aroma expression, cool-night grain filling and regional buyer demand are strong. It requires careful isolation from non-aromatic varieties during transplanting, harvest, drying and milling.

5.2 Basmati Variety

Basmati requires extra attention to grain length, varietal purity, lodging control, gentle milling and ageing. Maharashtra farmers should grow it mainly under contract or clear buyer specification because GI-linked authenticity and market acceptance are strict.

5.3 Kolam Variety

Kolam is suited to larger retail demand and has more flexible market acceptance than aromatic specialty rice. Grade value depends on whiteness, low broken percentage, uniform polish and clean storage.

5.4 Variety Performance Matrix

Variety	Yield Orientation	Aroma	Grain Quality	Best Districts	Special Pos
Indrayani	Medium	Strong regional aroma	Soft cooking, premium local	Pune, Satara, Kolhapur, parts of Konkan	Origin-linked aromatic rice.
Basmati	Medium under specialty management	Strong basmati aroma	Extra-long slender, elongates on cooking	Contract irrigated pockets	Premium long-grain market.
Kolam	Medium to good	Mild/neutral	Fine grain, daily cooking	Vidarbha, Konkan, western Maharashtra	High-volume retail

5.5 Yield, Aroma & Grain Quality Comparison

Quality Factor	Indrayani	Basmati	Kolam
Primary premium driver	Aroma and softness	Kernel length, purity and elongation	Fine grain and low broken
Milling care	Avoid overheating and aroma loss	Gentle milling to protect long kernels	Control polishing and broken rice
Admixture tolerance	Very low for premium pack	Very strict in standards-based trade	Moderate but affects grade
Ageing	Short controlled ageing improves aroma stability	Ageing often improves cooking quality	Usually sold fresh to moderately aged

6. SOIL REQUIREMENTS & PHYSICAL PROPERTIES

6.1 Ideal Soil Types

Soil Type	Suitability	Management Note
Clay loam / lowland clay	Excellent	Holds water for transplanted paddy; puddling effective.
Medium black soil	Good where drainage and water control exist	Avoid prolonged stagnation beyond crop need.
Lateritic Konkan soils	Good under bunded monsoon fields	Maintain organic matter and field bund integrity.
Sandy loam	Limited	Needs frequent irrigation and nutrient split; direct seeded risk is higher.

6.2 Soil pH & Organic Matter

Parameter	Ideal Range	Interpretation
pH	5.5–7.5	Rice tolerates mild acidity; extreme pH reduces nutrient availability.
Organic carbon	>0.5%	Improves soil structure, microbial activity and nitrogen buffering.
Available N	Low to medium common	Nitrogen is usually yield-limiting.
Zinc	Deficiency common in puddled/calcareous soils	Zn deficiency causes bronzing and stunting.
Iron toxicity	Possible in acidic lowlands	Use drainage and balanced nutrients where symptoms appear.

6.3 Land Preparation Protocol

- Repair bunds before monsoon to retain water and prevent nutrient loss.
- Plough and puddle transplanted fields to reduce percolation and suppress weeds.
- Level fields for uniform water depth, fertilizer distribution and crop maturity.
- In direct-seeded rice, prepare a fine, firm seedbed and manage early weeds aggressively.

7. CLIMATE REQUIREMENTS & WATER MANAGEMENT

7.1 Ideal Climate

Climate Parameter	Recommended Condition
Temperature	20–35°C; flowering is sensitive to extremes.
Rainfall	High-rainfall Konkan and eastern lowlands are naturally suited; irrigated pockets need assured water.
Humidity	High humidity favours rice growth but also blast and bacterial diseases.
Sunshine	Bright grain-filling weather improves yield and milling quality.

7.2 Water Requirement

Conventional transplanted rice is water-intensive because standing water, puddling and percolation losses are involved. Water productivity improves through field levelling, alternate wetting and drying, short-duration varieties and direct seeding where weed control is feasible.

7.3 Irrigation Management — Stage-wise Protocol

Stage	Water Management	Critical Priority
Nursery	Moist but not flooded immediately after sowing.	High
Establishment	Shallow water after transplanting; avoid seedling submergence.	High
Tillering	Maintain shallow standing water or AWD cycle.	Medium
Panicle initiation	Avoid water stress; this is a yield-critical stage.	Very High
Flowering	Maintain adequate moisture; stress causes sterility.	Very High
Ripening	Drain 7–10 days before harvest.	Medium

7.4 Water Saving Techniques

- Alternate wetting and drying in level fields with good bunds.
- System of Rice Intensification principles where labour and skill are available.
- Direct seeded rice in suitable soils with assured weed management.
- Laser or careful field levelling for uniform water depth.
- Short-duration varieties in water-limited command areas.

8. SOWING & TRANSPLANTING MANAGEMENT

8.1 Nursery Preparation

- Use clean seed, seed treatment and raised nursery beds in high-rainfall areas.
- Select fertile nursery soil with drainage to prevent seedling disease.
- Avoid over-aged seedlings; young vigorous seedlings establish better.

8.2 Direct Seeding Method

Direct seeded rice reduces labour and puddling requirement but demands precise sowing, early weed control and water management. It is better suited to level fields and farmers with timely herbicide/mechanical weeding access.

8.3 Transplanting Method

Transplanting remains dominant in Maharashtra paddy belts. Seedlings are transplanted into puddled fields with shallow water, followed by gap filling and early weed control.

8.4 Spacing Recommendations

System	Spacing	Seedling Age	Remarks
Conventional transplanted	20 × 15 cm or 20 × 20 cm	20–30 days	Adjust by variety duration and fertility.
SRI-style	25 × 25 cm	8–14 days	Single young seedling; needs skill and weeders.
Direct seeded	20 cm rows	Not applicable	Uniform seed drill depth and weed control essential.

8.5 Intercropping Systems

Rice is generally not intercropped in puddled lowlands. Bund crops such as pigeon pea, vegetables, fodder grasses or pulses may be used where bund width and water flow permit, but they should not compromise paddy water management.

9. SEED TREATMENT PROTOCOL

Treatment Step	Practice	Purpose
Salt-water selection	Remove floating, light and damaged seed; wash selected seed.	Improves seed vigour and uniformity.
Fungicide treatment	Use locally recommended fungicide as per label.	Controls seed-borne blast/brown spot and seedling rot.
Bio-inoculant	Use Azospirillum/PSB where recommended after chemical treatment dries.	Improves nutrient-use efficiency.
Soaking & incubation	Soak and pre-germinate seed for wet nursery where practiced.	Rapid and uniform nursery emergence.
Traceability entry	Record seed source, lot number, treatment and nursery date.	Creates audit-ready origin record.

Millets
Now

10. COMPLETE CROP CYCLE — STAGE-WISE GUIDE

Stage	Days	Operations	Quality Objective
Pre-season	May–June	Bund repair, soil test, seed procurement and nursery planning.	Verified field and seed readiness.
Nursery	0–25 days	Seed treatment, nursery sowing, water and disease care.	Vigorous seedlings.
Transplanting / sowing	20–35 DAS nursery	Puddling, levelling, transplanting or direct seeding.	Uniform stand.
Tillering	25–50 DAT	Weeding, nitrogen split, shallow water management.	High productive tillers.
Panicle initiation	45–65 DAT	Nutrient and water security; pest monitoring.	Strong panicle formation.
Flowering	70–90 DAT	Avoid water stress; monitor blast, BLB and stem borer.	Low sterility and healthy grain set.
Grain filling	90–115 DAT	Drain gradually; protect from lodging and birds.	High milling recovery.
Harvest	110–140 days	Harvest at 20–22% paddy moisture; dry safely.	Low breakage and safe storage.

Millets
Now

11. DISEASE & PEST MANAGEMENT

11.1 Blast Disease

Blast affects leaves, nodes and necks, especially under high humidity, susceptible varieties and excessive nitrogen. Neck blast directly reduces grain filling and quality.

11.2 Bacterial Leaf Blight

Bacterial leaf blight causes yellowing and drying from leaf tips and margins. It spreads through rain splash, wounds, floodwater and susceptible varieties.

11.3 Brown Spot

Brown spot is associated with nutrient stress, poor soil fertility and seed-borne infection. It reduces leaf area and can stain grains.

11.4 Stem Borer

Stem borer causes dead hearts at vegetative stage and white earheads at reproductive stage. Monitoring should begin early in tillering.

11.5 Leaf Folder

Leaf folder larvae fold leaves and scrape green tissue, reducing photosynthesis. Dense nitrogen-rich crops are more vulnerable.

11.6 Integrated Disease Management (IDM)

Problem	Stage	Symptoms	Integrated Management
Blast	Nursery to flowering	Leaf lesions, neck infection	Resistant variety, balanced N, seed treatment, advisory spray.
BLB	Tillering to grain filling	Tip drying and bacterial ooze	Clean seed, avoid clipping seedlings, drain field, resistant variety.
Brown spot	Nursery to maturity	Brown oval spots	Seed treatment, balanced nutrients, improve soil health.
Stem borer	Tillering to flowering	Dead hearts, white earheads	Light traps, egg mass removal, need-based control.
Leaf folder	Vegetative to booting	Folded scraped leaves	Avoid excess N, conserve natural enemies, need-based spray.

12. FERTILIZER MANAGEMENT — NPK, ORGANIC & MICRO-NUTRIENTS

12.1 NPK Recommendations (ICAR — Rice)

System	Nitrogen	Phosphorus	Potassium	Remarks
Rainfed lowland	60–80 kg N/ha	30–40 kg P ₂ O ₅ /ha	30–40 kg K ₂ O/ha	Adjust to soil test and rainfall.
Irrigated transplanted	100–120 kg N/ha	50–60 kg P ₂ O ₅ /ha	40–60 kg K ₂ O/ha	Split N; avoid excessive vegetative growth.
Premium aromatic	Moderate N	Soil-test P	Adequate K	Excess N can reduce aroma and increase lodging/disease.

12.2 Fertilizer Application Schedule

Timing	Application	Note
Basal	Full P, part K, and 25–50% N.	Incorporate before transplanting or final puddling.
Active tillering	Second N split.	Apply after weeding and with shallow water.
Panicle initiation	Final N and K split where recommended.	Critical for panicle size and grain filling.
Late season	Avoid unnecessary N.	Reduces lodging, disease and grain quality issues.

12.3 Organic Fertilizer Options

- FYM or compost before puddling to improve soil organic matter.
- Green manure with dhaincha/sunhemp where water and time permit.
- Rice straw incorporation with sufficient decomposition time.
- Azolla or blue-green algae in suitable standing-water systems.

12.4 Micronutrient Management

Micronutrient	Deficiency Symptom	Management
Zinc	Bronzing, stunting, poor tillering	Soil-test based zinc sulphate or approved foliar correction.
Iron	Chlorosis in upland/calcareous situations; toxicity in acidic lowlands	Correct drainage/pH and use advisory-based application.
Sulphur	Uniform yellowing of young leaves	Use sulphur-containing fertilizers where deficient.

13. GRAIN GRADING SYSTEM — A, B & C GRADES

13.1 Grade Parameters

Parameter	Grade A	Grade B	Grade C
Moisture	≤ 12–13% milled rice	≤ 14%	≤ 15% or re-dry required
Broken grains	Low, variety-specific premium limit	Moderate	High; lower market price
Foreign matter	Nil to trace	Low	Visible cleaning needed
Damaged/discoloured/chalky	Very low	Moderate	High
Varietal admixture	Very low; strict for aromatic lots	Controlled	Mixed lot
Odour/live insects	Absent	Absent	Reject or fumigate/reprocess

13.2 Grain Size & Moisture Standards

Rice grading uses grain length, breadth, broken percentage, fragments, chalkiness and moisture. Basmati requires long slender kernels and elongation after cooking, while Indrayani and Kolam require uniform grain type and low broken content for premium retail packs.

13.3 Aroma & Purity Standards

- Indrayani lots should be isolated by field, harvest and mill batch to protect aroma identity.
- Basmati requires strict non-basmati admixture control and grain-length verification.
- Kolam grade depends more on whiteness, polish consistency, broken rice and cleanliness.

13.4 Grade-wise Visual Characteristics

- Grade A: clean, uniform, bright grain with low broken and no stale odour.
- Grade B: acceptable grain with moderate broken or minor chalkiness.
- Grade C: mixed, high broken, dull or discoloured grain requiring reprocessing or discount sale.

13.5 Achieving Grade A — Key Practices

- Harvest at proper maturity and avoid over-drying before milling.
- Dry paddy slowly to safe moisture to reduce fissuring.
- Use clean threshing floors and avoid mixing varieties.
- Mill with correct rubber-roll settings and avoid overheating.
- Store rice in clean, dry, odour-free bags with batch and grade labels.

Quality Gate	Indrayani	Basmati	Kolam
Field isolation	Separate aromatic plots from non-aromatic paddy.	Maintain buyer-approved variety and seed source.	Avoid mixing coarse and fine paddy lots.
Drying check	Slow drying to protect aroma and reduce fissures.	Uniform drying before ageing and milling.	Dry evenly to reduce broken during milling.
Milling check	Moderate polish; preserve natural fragrance.	Gentle rubber-roll shelling; protect long kernels.	Consistent whiteness and polish level.

Quality Gate	Indrayani	Basmati	Kolam
Sample test	Aroma, moisture, broken and admixture.	Kernel length, elongation, aroma, broken and non-basmati grains.	Moisture, broken, chalky grains and brightness.
Pack approval	Origin label and lot QR only after grade pass.	Purity declaration only after assay confirmation.	Retail grade label after broken and moisture approval.

For premium rice, Grade A is not only a visual decision. It should combine representative sampling, moisture testing, varietal-purity review, milling recovery observation and pack-level traceability. A single failed quality gate should trigger re-cleaning, re-drying, downgrading or separate sale rather than blending into a premium lot.

Millets
Now

14. DISTRICT-WISE DEEP FARMING ANALYSIS — MAHARASHTRA

14.1 Pune District

Parameter	District Guidance
Climate / Soil	Western Ghats/Maval and irrigated valleys; strong Indrayani identity.
Best-fit Varieties	Indrayani and Kolam
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Maintain origin identity, aroma preservation and careful lot separation.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.2 Kolhapur District

Parameter	District Guidance
Climate / Soil	High rainfall and irrigation-supported paddy fields with good market access.
Best-fit Varieties	Indrayani-style aromatic, Kolam and local rice
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Disease monitoring under humid conditions and quality drying.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.3 Satara District

Parameter	District Guidance
Climate / Soil	Ghats, valleys and irrigated command pockets with specialty rice potential.
Best-fit Varieties	Indrayani, Kolam
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Use field levelling, bund repair and aroma-preserving post-harvest handling.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.4 Solapur District

Parameter	District Guidance
Climate / Soil	Water-limited district; rice only in assured irrigation pockets.
Best-fit Varieties	Kolam or contract specialty rice
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Strict water budgeting and short-duration varieties.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.5 Nashik District

Parameter	District Guidance
Climate / Soil	Selective irrigated pockets and tribal hill rice areas.
Best-fit Varieties	Kolam, local fine rice, limited aromatic lots
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Match variety to water source and market channel.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.6 Nagpur District

Parameter	District Guidance
Climate / Soil	Vidarbha paddy pockets with milling and retail linkages.
Best-fit Varieties	Kolam and fine rice
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Focus on grain brightness, low broken and warehouse hygiene.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.7 Chandrapur District

Parameter	District Guidance
Climate / Soil	Traditional eastern Maharashtra lowland rice zone.
Best-fit Varieties	Kolam, fine rice and local paddy
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Manage blast/BLB and harvest timing under humid conditions.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.8 Gondia District

Parameter	District Guidance
Climate / Soil	Major paddy district with lowland fields and strong rice culture.
Best-fit Varieties	Kolam and other fine rice
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Cluster-level grading, drying and mill integration.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.9 Bhandara District

Parameter	District Guidance
Climate / Soil	Tank/canal-influenced paddy belt with high rice dependence.
Best-fit Varieties	Kolam and fine/medium rice
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Water control and post-harvest quality management.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.10 Raigad District

Parameter	District Guidance
Climate / Soil	Konkan high rainfall, bunded fields and local rice consumption.
Best-fit Varieties	Kolam, local fine rice, selected aromatic
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Drainage after heavy rain and safe drying in humid weather.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.11 Ratnagiri District

Parameter	District Guidance
Climate / Soil	Coastal lateritic fields with monsoon-dependent paddy.
Best-fit Varieties	Kolam/local rice and aromatic niches
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Bund integrity, disease care and moisture-safe storage.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

14.12 Sindhudurg District

Parameter	District Guidance
Climate / Soil	High rainfall coastal and Ghats paddy ecology.
Best-fit Varieties	Kolam/local fine rice
Water Management	Maintain bunds, shallow standing water at establishment, and drain before harvest.
Disease Focus	Blast, bacterial leaf blight and sheath/leaf diseases in humid periods.
Grading & Market Note	Small-lot traceability, clean drying and local branding.

- Record variety, nursery date, transplanting date and harvest date for every lot.
- Keep aromatic and non-aromatic paddy separate during threshing, drying and milling.
- Use moisture testing before bagging to prevent storage odour and fungal risk.

Millets
Now

15. HARVESTING, THRESHING & STORAGE

15.1 Harvesting Guidelines

- Harvest when 80–85% grains in panicles are mature and straw begins yellowing.
- Avoid late harvest, which increases shattering, lodging, fissuring and pest damage.
- Keep varieties and fields separate at harvest for traceability.

15.2 Threshing

- Thresh on clean surface or use mechanical threshers adjusted to reduce grain injury.
- Remove straw, chaff and immature grains before drying.

15.3 Drying

Dry paddy gradually to about 14% for safe storage and lower before milling as required by buyer specification. Rapid high-temperature drying can fissure kernels and increase broken rice during milling.

15.4 Storage Best Practices

Storage Factor	Best Practice
Moisture	Safe dry paddy/rice; verify before bagging.
Bags	Clean, dry and labelled with lot ID, variety and date.
Warehouse	Pallets, ventilation, wall clearance and rodent control.
Aroma protection	Avoid storage near chemicals, spices, diesel or odorous material.
Inspection	Monitor monthly for insects, moisture migration and stale smell.

16. INPUT CYCLE & COST STRUCTURE

16.1 Complete Input Cycle

Input / Operation	Timing	Typical Requirement	Quality Impact
Seed	Before nursery	Certified Indrayani/Basmati/Kolam seed	Varietal purity and uniform maturity.
Nursery	Pre-transplant	Seedbed, treatment, water and disease care	Strong seedlings.
Land prep	Before transplanting	Bund repair, puddling, levelling	Uniform water and nutrient use.
Fertilizer	Basal and splits	NPK as per soil test	Yield, aroma and grain filling.
Weed control	Early crop	Manual, cono-weeder or herbicide as advised	Tillering and yield protection.
Plant protection	Need-based	Monitoring and advisory control	Avoids grain staining and yield loss.
Harvest/milling	Maturity onward	Clean threshing, drying and milling	Broken control and grade.

16.2 Cost Analysis per Acre

Cost Head	Relative Cost Level	Management Note
Seed and nursery	Low to medium	Certified seed cost is justified for premium traceability.
Land preparation/puddling	High	Fuel, labour and water availability drive cost.
Fertilizer and micronutrients	Medium	Soil-test based application avoids waste.
Labour for transplanting/weeding	High	Mechanisation or DSR can reduce labour where suitable.
Harvesting, drying and milling	Medium to high	Quality handling improves grade premium.
Traceability and grading	Low	Moisture testing, tags and records add buyer confidence.

17. MILLETS NOW — DIGITAL TRACEABILITY PLATFORM (V2)

17.1 Platform Vision & Goals

- Create lot-level proof of origin, variety, crop practice and grade for premium rice.
- Protect Indrayani aroma identity, Basmati purity and Kolam grade consistency.
- Support farmer groups, mills and consumers with QR-based verification.

17.2 End-to-End Traceability Workflow

- Farmer registration and plot geo-tagging.
- Seed lot, nursery date, transplanting/direct seeding record.
- Crop-stage input, irrigation, pest and disease monitoring.
- Harvest, threshing, drying, paddy storage and milling batch entry.
- Grading, packing, QR code generation and buyer dispatch.

17.3 System Architecture Layers

Layer	Components
Farmer Application	Farm profile, plot map, variety, nursery and input records.
Field Officer Console	Inspection checklist, photo evidence, disease notes and grade approval.
Mill Layer	Paddy receipt, drying, milling batch, broken percentage and polish status.
Quality Layer	Moisture, broken, chalky, damaged, foreign matter, aroma and purity records.
Blockchain Layer	Lot hash, custody transfer, timestamp and immutable verification record.
Consumer Interface	QR scan, farmer origin, variety, grade, milling date and cooking notes.

17.4 Blockchain Traceability Record — Data Fields

Data Field	Example Entry
Crop / Variety	Rice / Indrayani
Farmer ID / FPO ID	MN-RC-PUN-000218
Plot and district	Geo-tagged plot, Pune district
Seed lot	Certified Indrayani seed lot number
Nursery and transplanting date	12 June 2026 / 05 July 2026
Harvest and milling date	28 October 2026 / 15 November 2026
Grade	Grade A, aromatic, low broken
Bag / pack IDs	QR pack series MN-RC-2026-0001 to 0500

17.5 QR Code Based Farmer-to-Consumer Traceability

Each retail pack can carry a QR code showing farmer/FPO, district, variety, crop year, milling date, grade parameters and authenticity status. This is especially valuable for Indrayani and Basmati where

admixture risk is high.

Millets
Now

18. RICE VALUE ADDITION & MARKET PRODUCTS

18.1 Traditional Rice Products

- Steamed rice, bhat, varan-bhat, curd rice, festive rice dishes and local Konkan preparations.
- Flattened rice/poha in suitable processing chains.
- Hand-pounded or semi-polished rice for traditional consumers.

18.2 Premium Rice Products

- Indrayani origin-labelled aromatic rice packs.
- Aged Basmati-style long-grain premium packs.
- Fine Kolam retail packs for household daily use.

18.3 Rice Flour & Snacks

- Rice flour for bhakri, modak, dosa/idli blends and bakery mixes.
- Papad, chakli, extruded snacks and instant mixes.
- Gluten-free flour blends for health-focused markets.

18.4 Rice Bran Oil

Rice bran from milling contains oil and nutraceutical fractions. Stabilisation of bran soon after milling is essential to prevent rancidity and preserve oil quality.

18.5 Industrial Uses

Use	Input Form	Market Note
Starch	Broken rice / flour	Food, textile and adhesive applications.
Brewing / fermentation	Broken rice	Industrial fermentation and beverage use where permitted.
Animal feed	Bran and broken rice	Value recovery from milling by-products.
Ready-to-cook foods	Flour, flakes, parboiled rice	Urban convenience segment.

19. GOVERNMENT SCHEMES & POLICY SUPPORT

19.1 Key National Schemes for Rice Farmers

Scheme / Institution	Support Relevance
ICAR-IIRR / AICRIP	Rice research, varieties, production technology and disease management.
National Food Security Mission	Seed, demonstrations and productivity improvement support.
PM-KISAN / farmer support schemes	Income support for eligible farmers.
PMFME	Micro food processing support for rice products, flour and snacks.
e-NAM / assaying systems	Quality-based market linkage and transparent transactions.

19.2 Maharashtra State Support

- State agriculture advisories for district-wise paddy cultivation and plant protection.
- KVK demonstrations on improved varieties, SRI, direct seeding and integrated nutrient management.
- FPO and warehouse support for aggregation, grading, milling and retail linkage.

19.3 Crop Insurance & MSP

Rice farmers should check current-season eligibility under crop insurance, notified areas, cut-off dates and premium details. Minimum Support Price procurement applies as per Government of India and state procurement arrangements for the relevant marketing season.

Minister
Now

20. QUICK REFERENCE SUMMARY TABLE

Parameter	Rice Recommendation
Crop	Rice / Paddy
Botanical name	Oryza sativa L.
Focus varieties	Indrayani, Basmati and Kolam
Districts	Pune, Kolhapur, Satara, Solapur, Nashik, Nagpur, Chandrapur, Gondia, Bhandara, Raigad, Ratnagiri, Sindhudurg
Season	Primarily kharif; summer/rabi only under assured irrigation.
Soil	Clay loam, lowland clay and banded lateritic paddy soils.
pH	5.5–7.5 preferred.
Water	Standing water or AWD depending on method and field control.
Critical stages	Establishment, panicle initiation, flowering and grain filling.
Major diseases/pests	Blast, BLB, brown spot, stem borer and leaf folder.
Grade A focus	Low moisture, low broken, purity, aroma/whiteness and no odour.
Traceability capture	Seed lot, field, variety, crop practices, harvest, milling batch, grade and QR pack ID.

— END OF REPORT —

Millet
Now